

New Local Administrator Creation Detection and Investigation using Windows Security Logs and Splunk

Objective

The objective of this investigation was to analyze Windows Security events generated during the creation of a new local user account and the subsequent assignment of administrative privileges. The investigation focused on identifying account creation, account modification, password initialization, group membership changes, and privilege escalation using Splunk.

MITRE ATT&CK Mapping

Tactic	Technique	ID
Persistence	Create Account: Local Account	T1136.001
Privilege Escalation	Account Manipulation	T1098
Privilege Escalation	Additional Local Groups	T1098.007

Lab Environment

Component	Technology
SIEM	Splunk Enterprise
Endpoint	Windows 11
Log Source	Windows Security Logs
Log Forwarder	Splunk Universal Forwarder

Attack Simulation

The following commands were executed from an elevated Command Prompt.

```
net user eviladmin P@ssw0rd123! /add
```

```
net localgroup administrators eviladmin /add
```

Attack Chain Overview

Step	Activity
1	Local user account created
2	Account enabled
3	Password initialized
4	Account properties updated
5	User added to default Users group
6	User added to local Administrators group
7	Group membership verified

Key Windows Security Events

Event ID	Description
4720	User account created
4722	User account enabled
4724	Password reset
4738	User account modified
4732	Added to local security group
4798	Local group membership enumerated

Investigation

Event ID 4720 – Local User Account Creation

Key Findings

Field	Value
Event ID	4720
Created Account	eviladmin
Created By	socadmin
Host	win-11
Result	Success

Analysis

Windows Security Event ID 4720 confirmed that a new local account named **eviladmin** was created by the administrative user **socadmin**.

This event establishes the initial account creation and serves as the starting point of the investigation timeline.

Event ID 4722 – Account Enabled

Key Findings

Field	Value
Event ID	4722
Account	eviladmin
Performed By	socadmin

Analysis

Immediately after creation, the newly created account was enabled, making it available for authentication.

This indicates the account was prepared for operational use rather than being left in a disabled state.

Event ID 4724 – Password Initialization

Key Findings

Field	Value
Event ID	4724

Field	Value
Target Account	eviladmin
Performed By	socadmin

Analysis

Windows recorded a password reset event for the newly created account.

This confirms that valid credentials were assigned before the account was used.

Event ID 4738 – Account Modification

Key Findings

Field	Value
Event ID	4738
Account	eviladmin
Old UAC	0x15
New UAC	0x10

Analysis

The account properties were updated after creation.

The User Account Control attributes changed from a disabled account to an enabled account with a configured password.

Event ID 4732 – Added to Users Group

Key Findings

Field	Value
Event ID	4732
Group	Users
Member	eviladmin
Added By	socadmin

Analysis

The newly created account was automatically added to the default local **Users** group. This behavior is expected during local account creation.

Event ID 4732 – Added to Administrators Group

Key Findings

Field	Value
Event ID	4732
Group	Administrators
Member	eviladmin
Added By	socadmin

Analysis

The account **eviladmin** was added to the local **Administrators** group. This represents the privilege escalation stage of the attack simulation and provides the highest-confidence indicator within this investigation. Creation of privileged local accounts should always be reviewed against approved administrative activity.

Event ID 4798 – Group Membership Enumeration

Key Findings

Field	Value
Event ID	4798
Account	eviladmin
Process	net1.exe

Analysis

The account's local group memberships were enumerated using **net1.exe**. This verified that the account had been successfully assigned its intended group memberships.

Splunk Investigation

Detect User Account Creation

```
index=windows EventCode=4720
```

Detect User Account Enablement

```
index=windows EventCode=4722
```

Detect Password Reset

```
index=windows EventCode=4724
```

Detect Account Modification

```
index=windows EventCode=4738
```

Detect Group Membership Changes

```
index=windows EventCode=4732
```

Detect Group Membership Enumeration

```
index=windows EventCode=4798
```

Reconstruct the Attack Timeline

```
index=windows Target_Account_Name=eviladmin  
| sort _time
```

```
| table _time EventCode Subject_Account_Name Target_Account_Name Group_Name  
Process_Name Message
```

Indicators of Compromise (IOCs)

Indicator	Value
Username	eviladmin
Creator Account	socadmin
Elevated Group	Administrators
Host	win-11
Enumeration Tool	net1.exe

Timeline

Time	Event
23:13:05	User account created
23:13:05	Account enabled
23:13:05	Password initialized
23:13:05	Account properties modified
23:13:05	Added to Users group
23:13:13	Added to Administrators group
23:13:22	Group membership enumerated

Detection Opportunities

- Monitor Event ID **4720** for newly created local accounts.
 - Correlate Event ID **4732** when accounts are added to the **Administrators** group.
 - Alert when account creation and administrator group assignment occur within a short time window.
 - Investigate newly created administrator accounts created outside approved maintenance windows.
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Analyst Conclusion

A new local user account named **eviladmin** was created by **socadmin** on the Windows endpoint. The account was subsequently enabled, assigned a password, modified, and added to both the default **Users** group and the local **Administrators** group. Finally, the account's group memberships were verified using **net1.exe**. The correlated Windows Security events provide a complete audit trail of account creation and privilege assignment. While this activity was performed in a controlled lab environment, the same behavioral pattern is commonly associated with attacker persistence and privilege escalation and should be treated as high priority during SOC investigations.